

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question

$2x - y > 883$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

Answer

A.

x	y
440	0
441	-2
442	-4

B.

x	y
440	0
442	-2
441	-4

C.

x	y
442	0
440	-2
441	-4

D.

x	y
442	0
441	-2
440	-4

Correct Answer: D

Rationale

Choice D is correct. All the tables in the choices have the same three values of x , **440**, **441**, and **442**, so each of the three values of x can be substituted in the given inequality to compare the corresponding values of y in each of the tables. Substituting **440** for x in the given inequality yields $2(440) - y > 883$, or $880 - y > 883$. Subtracting **880** from both sides of this inequality yields $-y > 3$. Dividing both sides of this inequality by -1 yields $y < -3$. Therefore, when $x = 440$, the corresponding value of y must be less than -3 . Substituting **441** for x in the given inequality yields $2(441) - y > 883$, or $882 - y > 883$. Subtracting **882** from both sides of this inequality yields $-y > 1$. Dividing both sides of this inequality by -1 yields $y < -1$. Therefore, when $x = 441$, the corresponding value of y must be less than -1 . Substituting **442** for x in the given inequality yields $2(442) - y > 883$, or $884 - y > 883$. Subtracting **884** from both sides of this inequality yields $-y > -1$. Dividing both sides of this inequality by -1 yields $y < 1$. Therefore, when $x = 442$, the corresponding value of y must be less than **1**. For the table in choice D, when $x = 440$, the corresponding value of y is -4 , which is less than -3 ; when $x = 441$, the corresponding value of y is -2 , which is less than -1 ; when $x = 442$, the corresponding value of y is **0**, which is less than **1**. Therefore, the table in choice D gives values of x and their corresponding values of y that are all solutions to the given inequality.

Choice A is incorrect. When $x = 440$, the corresponding value of y in this table is **0**, which isn't less than -3 .

Choice B is incorrect. When $x = 440$, the corresponding value of y in this table is 0 , which isn't less than -3 .

Choice C is incorrect. When $x = 440$, the corresponding value of y in this table is -2 , which isn't less than -3 .